Time : 3 hours

Note : *i) Answer* ***Question No. 1*** *(Compulsory) and answer any* ***four questions*** *remaining questions (2- 7). Each question caries 12 marks.*

ii) Answers must be written in same order as they occur in the Question Pa

iii) ***Missing data, if any may suitably be assumed.***

Max. Ma

Mark

1. a) What advantage is there in having different time-quantum sizes at different levels of a multilevel queuing system. 2

b) Why are page sizes always power of 2? 2

c) When is a process said to be in the resident set? 2

d) Explain the purpose of the open() and close() operations. 2

e) Capability lists are usually kept within the address space of the user. How does the system ensure that the user cannot modify the contents of the list? 2

f) Sketch the layers in a Linux system. 2

2. a) One set of algorithms may include another (for example, the FCFS algorithm is the RR algorithm with an infinite time quantum). What (if any) relation holds between the following pairs of algorithm sets? 7

i) Priority and SJF

ii) Multilevel feedback queues and FCFS

iii) Priority and FCFS

iv) RR and SJF

b) What are the necessary conditions for deadlocks to occur? How is deadlock prevention related? Justify your answers with examples. 5

3. a) If FIFO page replacement is used with four page frames and eight pages, how many page faults will occur with the reference string 0172327103 if the four frames are initially empty? Now repeat this problem for LRU. 6

b) What are the two difficulties with the use of equal-size fixed partitions? 6

4. a) Summarize three techniques for performing I/O and give relationship among those. 6

b) Discuss in detail on file system software architecture. 6

5. a) The assets of a computer system can be categorized as hardware, software, data, and communication lines and networks. Briefly describe these four categories and relate these to the concepts of integrity, confidentiality, and availability. 6

b) Write a short notes on Token based authentication, password based authentication. 6

6. a) Discuss three advantages of dynamic (shared) linkage of libraries compared with static linkage. Describe two cases in which static linkage is preferable. 6

b) Explain WIN32 Application Programming Interface. 6

7. a) Explain the use of segmentation and mention the steps needed for address translation. 6

b) What is Thrashing? What are the techniques to overcome the problem? 6

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